Xilin Mao

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Education

Tsinghua University, Qiuzhen College, Ph.D. in Statistics — Sep 2024 – Jun 2029 (expected)

- Advisor: Prof. Yuhong Yang
- Research interests: statistical machine learning, causal inference, and model selection and aggregation
- Current GPA: 3.80/4.00

University of Chinese Academy of Sciences, School of Mathematical Sciences, B.S. in Mathematics — Sep 2020 – Jul 2024

- GPA: 3.87/4.00

Research Experience

Model Aggregation for Policy Learning — May 2026 – Present

- Developing model-aggregation and exponential-weighting methods for statistical policy learning, including settings with unmeasured confounding.
- Study theoretical performance guarantees for aggregated decision rules and evaluate the proposed methods through simulation experiments.
- Methods include statistical decision theory, model aggregation, machine learning, and causal inference.

Data Combination in Causal Inference — *Manuscript submitted to JASA* — May 2025 – Aug 2026

- Developed efficient estimators for differences in conditional average treatment effects (CATEs) between randomized trials and observational studies in the presence of unmeasured confounding.
- Derived theoretical properties of the proposed estimators and developed sensitivity-analysis procedures for violations of causal identification assumptions.
- Used semiparametric inference, M-estimation, empirical-process theory, and simulation, real data studies to establish and evaluate statistical performance.

Quantitative Finance Project

Alpha Factor Mining with Large Language Models — Apr 2026 – Jun 2026

- Surveyed recent methods for using large language models to generate and discover quantitative trading factors, and reproduced two representative research papers.
- Proposed an improved factor-mining algorithm based on findings from the reproduction studies.

Additional Research

Homotopy Theory in Complex Geometry — Nov 2023 – May 2024

- Undergraduate thesis supervised by Prof. Songyan Xie; studied model-category structures and Oka theory from a homotopy-theoretic perspective.

Technical Skills

Programming: Python, R; working knowledge of C and MATLAB

Statistical Methods: machine learning, causal inference, semiparametric inference, empirical processes, model selection and aggregation

Awards and Honors

Bronze Medal, Team Competition, S.-T. Yau College Student Mathematics Contest, Yau Mathematical Sciences Center, Tsinghua University — Jun 2023

First-Class Academic Scholarship, University of Chinese Academy of Sciences — Nov 2022